Solutions to Selected Problems

Problems of Chap. 1

1.3 Medial.

1.13 A football, because of how it needs to be held.

1.14 Head.

1.27 (a) (partial answer) For lower legs 3.72–9.30 kg and 5.53–9.55 kg.

1.32 (b) 0.25 m and 0.50 m.

1.33 (partial answer) Surface area is 20.1 sq ft.

1.42 (b) 24.0 kg/m^2 for 1.68 m and 26.1 kg/m^2 for 1.8 m; (c) yes.

1.47 (a) 92.6 kg (204.2 lb) for Man A and 84.3 kg (185.9 lb) for Man B; (b) 0.4 kg (0.9 lb) for Man A and 8.7 kg (19.2 lb) for Man B.

1.52 (b) 21.1 and 27.0.

1.60 Bigger in cold climate.

© Springer International Publishing Switzerland 2016

I.P. Herman, *Physics of the Human Body*, Biological and Medical Physics, Biomedical Engineering, DOI 10.1007/978-3-319-23932-3

Problems of Chap. 2

2.1 Third class lever.

2.2 Triceps brachii, second class lever.

2.3 (a) First; (b) second; (c) third class levers.

2.11 (b) $m_{\text{leg}}(x_{\text{extended leg}} + x_{\text{balancing leg}}) = (m_{\text{torso+head}} + 2m_{\text{arm}})x_{\text{upper body}}$.

2.13 (a) $-m_{\text{b}}x$.

2.14 $T_1 = W_1 = 223.6$ N, $T_2 = W_2 = 282.8$ N, $\alpha = 26.6°$.

2.15 (partial answer) 428 N at an angle of 0.073 rad above the horizontal.

2.18 (b) 310.6 N.

2.22 (b) $T = 131$ lb, $F = 208$ lb, angle of **F** is 29.8°.

2.34 (partial answer) −6 N-m for 20 cm deep.

2.35 (partial answer) −6.9 N-m for upright, −19.25 N-m for bent.

2.36 (partial answer) −21.25 N-m for bent over, with bent knees and the object far from her body.

Problems of Chap. 3

3.7 Children have proportionately larger heads and shorter legs.

3.13 0.27, easy to achieve with cleated running shoes.

3.17 1.7127 kg-m^2.

3.21 Activate either muscle 3 or muscles 1, 3, and 5 with muscle 1 being activated enough to cancel the torque on the hip by muscle 5. The advantage of the latter arrangement is that greater net torque can be applied to the knee. Activation of additional muscle groups is possible, but their net torques would cancel and this would be of no value.

3.39 2/3.

3.46 (partial answer) (a) No, they are defined differently.

3.52 Yes, by $-0.07\,\text{m}$, yes.

3.53 For (a) $0.254H$; (b) $0.150H$; (c) $0.077H$; (d) $0.150H$; (e) $0.254H$; 14 cm.

3.57 Yes, because how fast the body can take off at a given angle depends on the construction of the body's feet and legs.

3.59 It improves their stability by increasing their moments of inertia (so for a given torque their angular acceleration will be less) and lowers the centers of mass of the combinations of perfomer and pole.

3.62 Shoulder joint: horizontal abduction; shoulder girdle: adduction; elbows: flexion; wrists: reduction of hypertension; head and neck: none; lumbar spine: none; hips, none.

3.71 (a) 25%; (b) 3.5× her body weight.

3.72 (a) With the origin at the base of the tower, the slope is characterized by $y = -x$ and at time t the skier is at $x = (15\,\text{m/s})t$, $y = 10 - (5\,\text{m/s}^2)t^2$. The skier lands when these curves intersect, with $-(15\,\text{m/s})t = 10 - (5\,\text{m/s}^2)t^2$, which occurs at $t = 3.56\,\text{s}$. Then $x = -y = 53.4\,\text{m}$ and the distance is $\sqrt{x^2 + y^2} = 75.5\,\text{m}$ [157]. (b) With drag and lift included, this distance is 65.9 m, so the effect of drag dominates that of lift and it is significant.

3.86 2.16 kg.

3.87 (a) 3,880 W; (b) ∼3.6–7.2×; (c) ∼2.1–4.1×; (d) The shoulder muscles apparently cannot supply all of this kinetic power, so recovery of elastic energy stored in shoulder tendons and ligaments during an earlier stage of the pitching motion may also be important.

3.88 Roughly 12.5–13.5 psi, roughly 31.2–33.7 pounds.

3.89 (b) (partial answer) Estimating a football volume of $200\,\text{in}^3$ and using the mass density of air at atmospheric pressure (14.7 psi) of $\sim 1.2\,\text{kg/m}^3$ at ∼70 °F, a decrease in pressure by 2 psi from 12.5 psi, would decrease the weight of the air in the ball by 0.02 oz, which is much less the allowable 1 oz variation in ball weight.

3.94 (partial answer) (a) $\alpha_2 = 90° - \alpha_1$.

3.95 (a) (partial answer) $\sin\alpha = u/v$. For shallower angles the ball travels farther, but the hang time is so short that the cover man cannot arrive when the punt returner catches the ball (who can then elude the defender and return the ball) and for steeper angles the hang time is long enough to enable the cover man to reach the punt returner in time, but the punt distance is too short.
(b) 69.8°, 43.7 yards, 4.70 s, compared to the results optimizing the range with 45°: 67.4 yds, 3.54 s. Drag greatly changes these results [135].

3.104 The initial conditions give $b_1 = 17.1$, leading to $b_2 = 1.10$ and $b_3 = 1.56$, and to 47.3 m range and 3.72 s travel time.

3.116 The velocity and spin vectors point in the same direction.

3.120 44.0°, 31.1 m/s.

3.125 3.13, 3.04, and 3.27 m/s and 14, −4, and −22° for backspin of 100, 200 and 300 rad/s.

3.127 $(3/5)v_0$.

3.136 (partial answer) Elastic collisions are likely fatal for collision times <120 ms.

3.142 (a) 150 ft/s^2, 4.7 g, (b) 1,150 lb.

3.146 (partial answer) (c) The model predictions are not very close to the observed values, but the inelastic collison model is better because only it agrees with the observation that it is easier to break the given wooden board.

3.147 (partial answer) Both the inelastic and collision models do not agree with the observation that it is easier to break the given wooden board, so it is better to use the mass of the hand in the model, as in Problem 3.146, than of the hand and forearm combination.

3.149 The momentum transfer when firing a bullet or being hit by one is equal in magnitude to that of the bullet, but in a direction opposite to the bullet (Newton's Third Law), so recoil is expected from the shooter also. Aside from reflex action, there is some recoil expected when the bullet hits the person. For a ∼0.03 kg bullet traveling at ∼300 m/s stopping in a 70 kg person, recoil should occur at a rate of ∼0.13 m/s or ∼5 inches/s. This treats the whole body as a rigid body that is not touching the ground.

3.150 (a) 1.4 m/s; (b) 0.42 ms; (c) 930 s; (d) likelihood of death due to brain acceleration is about 50%, but of course one needs to consider the physical damage caused by bullet if it enters the brain, and so on.

3.152 See [260] from Chap. 3.

3.154 e^2.

3.155 1.44, 1.06, 1.01, and 0.18 m.

3.156 0.50, 0.53, and 0.55.

3.163 48.0 mph before, 24.7 mph after.

3.164 40.5 oz and about 80 mph.

3.178 413 N.

Problems of Chap. 4

4.2 (c) 480 MPa.

4.5 (a) 30 Pa, (b) 67 mm^2, (c) 1%.

4.6 (partial answer) 1.6 MPa for nails.

4.7 (partial answer) 0.0031 for nails.

4.8 (partial answer) 8,000 N/m^3 for nails.

4.9 (partial answer) 780 N/m^3 for nails.

4.15 (a) Tension.

4.20 (a) $\lambda = 2$, $\epsilon_{\text{small}} = 1$, $\epsilon_{\text{general}} = 3/2$.

4.21 (a) $\lambda = 2$, $\epsilon_{\text{small}} = 1$, $e = 3/8$.

4.22 (a) $\lambda = 2$, $\epsilon_{\text{small}} = 1$, $\ln \lambda = 0.69$.

4.23 (a) Yes, because the dashpot resistive force increases with speed;
(b) Yes, because the spring supplies the needed restoring force to return it to its equilibrium position.

4.24 Length is 3.1 cm, $\mathrm{d}x/\mathrm{d}t$ is 0.05 cm/s.

4.28 (a) $2\theta(t+1) - 2\theta(t-3)$, with all times in seconds.

4.33 It becomes the Voigt model, the Maxwell model, and a dashpot, respectively.

4.42 (a) 6,900 N.

Problems of Chap. 5

5.3 428 N, 0.07 rad above the horizontal.

5.9 (a) 0.11, it is larger than the 0.09 listed in the table—but in linear theory it would be expected to be UTS$/Y = 0.22$, (b) $fY = 135\,\mathrm{N/cm^2}$, (c) the diameter of the tendons is 0.073× that of the muscle, (d) $2{,}800\,\mathrm{N/cm^2} = 28\,\mathrm{MPa}$, which is less than the 54 MPa UTS listed in Table 4.2, so it is less than it, even with linear theory.

5.11 (a) ∼3,880 W/kg; (b) The increase in kinetic energy per unit time and mass of shoulder muscle is ∼1,800 W/kg is ∼2–4× larger than the maximum power derivable from pre-stretched muscle, 438–875 W/kg, when used perfectly efficiently; (c) As noted in Chap. 3 and [408], the extra power needed to drive humerus rotation may come from the recovery of elastic energy stored in shoulder tendons and ligaments during the cocking phase of the pitch.

5.13 That the contraction times are unaffected by the pre-stretching.

5.15 The muscles are fairly near their optimal lengths. However, there are significant changes in the lengths during bicycling, but less than the maximum expected for muscles are shown for several reasons: (a) The decreases in contracted muscle length are actually greater than those shown because tendon extension will lessen the decrease in the (plotted) total muscle/tendon length, (b) the bicycle is set to use muscles in their optimal state, both in muscle length and speed, so the muscles will not be much longer or shorter than their optimal length.

5.17 (a) (partial answer) $\sqrt{3}N F_{fiber}/2$.

5.27 (a) 40 s.

5.36 (b) 4000 sarcomeres in series × 3200 nm/s per sarcomere = 1.28 cm/s.

5.39 (a) (partial answer) 2,770 W.

5.41 (d) (partial answer) They need to be a little greater during flexion because you are causing motion and not maintaining a static equilibrium condition.

5.42 (c) (partial answer) Yes. With constant load training, the muscles will meet much resistance only for muscle lengths smaller and larger than L_0 and little resistance near L_0 (and the corresponding joint angles). With variable resistance training, the load can be set to provide resistance over a wide range of muscle lengths and joint angles (for that particular joint).

Problems of Chap. 6

6.1 (partial answer) 0.3°C.

6.4 (b) 144 BTU.

6.6 (a) (partial answer) 7.1 kcal/g.

6.14 (a) 261 g, compared to 260 g and 280 g; (b) 45%; (c) 18%.

6.19 90% of the fruit and 40% of the dried fruit is water plus non-metabolizable matter.

6.26 (b) 31.4 kcal/g, which is 11% too high; (c) 36.25 kcal/g, which is 28% too too high.

6.27 2,200 kcal.

6.32 (a) −12.6 kcal/mol; (b) −14.0 kcal/mol.

6.33 66–70%.

6.38 (a) 70 kg; (b) This is somewhat larger than the body mass of 60 kg; (c) 700 cycles/day, 0.5 cycles/min.

6.43 147 moles of ATP.

6.49 6.9.

6.55 (a) 321.4; (b) 1.15; (c) 2.3 standard donuts.

6.62 (partial answer) 580 kcal/h for 50 kg college-age women, using 40 mL/kg-min and a calorific equivalent of 4.83 kcal/L O_2.

6.63 (a) 89.9 m/min, 1.50 m/s, 3.35 mph, very good agreement; (b) 259 kcal/hr.

6.64 ~3000 kcal.

6.68 (partial answer) (a) 1048 kcal/h and 3.6 L O_2/min for Stage VIII, (b) 35 kcal for Stage VIII.

6.73 (a) 305 J, (b) 36 kcal, (c) no, but it excludes the pushing for 5–10 s during each play, which obviously accounts for most of the energy expenditure (although relatively little of the work done).

6.76 12.2 kJ, 11.7 kcal.

6.80 From the change in potential energy only the minimum would be 0.9 kcal.

6.85 Activity factor is 1.48 (assuming "self-care" walking and also cycling at 5.5 mph), MR is 1920 kcal/day (using 1300 kcal/day BMR).

6.86 (a) 4.9, (b) 4.2.

6.90 Assuming 9 kcal/g fat gives a gain of 3.2 kg, which is consistent with the observations.

6.92 15 kg, 33 lb.

6.100 2.2 L/h.

6.106 Thermal losses to the outside world are smaller with mittens. Except for the outermost ones, the lateral sides of the four fingers in the mittens are in contact with each other, rather than the material and then the outside world, so thermal losses are smaller.

6.107 (a) 70 kcal/h.

6.112 $T_{\mathrm{wc}} = 35.74 + 0.6215T - 35.75w^{0.16} + 0.4275Tw^{0.16}$.

6.113 (a) A and B; (b) C; (c) A.

6.118 The first term is 40 kcal/day, second term in 220 kcal/day. The second term is very significant.

Problems of Chap. 7

7.3 2,240 cm^3, 2.49 kg, 24.4 N; using a mass density of 1.11 g/cm^3.

7.7 No, his density is then 1.01 g/cm^3, which is above that of water, using a fat density of 0.8 g/cm^3.

7.18 (a) Q/N; (b) The diameter of the small tubes would then be $N^{1/4}D$, which is not possible because the diameter of the small tubes would then exceed that of the larger tubes.

7.25 (partial answer) 0.32 cm in a gas.

7.39 $Re = 0.001$, viscous/laminar.

Problems of Chap. 8

8.5 You should be concerned, but not about your blood pressure (which is really 120 mmHg/80 mmHg), but about the person who told you your blood pressure in absolute pressure instead of the standard gauge pressure.

8.9 1.16× larger

8.13 (a) 121 mmHg, which is 32% higher than the 92 mmHg base case, (b) 115 mmHg, which is 25% higher.

8.16 (a),(b) They change by a factor of 1/8.

8.20 1 N-s/m^5 = 10^6 (N/m^2)/(cm^3/s) = 10^5 dyne-s/cm^5 = 1.32×10^8 PRU.

8.32 $u_1/4$.

8.33 (partial answer) 0.012/cm for the subclavian artery

8.40 98.6 mmHg, using a blood density of 1060 kg/m^3.

8.45 Type I skeletal muscle, because very fast response is not needed and endurance is essential.

8.49 0.128 L = 128 cm^3.

8.50 (partial answer) 3.6 cm inner radius.

8.66 Heart beat rate: 83/min for the man, 91/min for the woman, 161/min for the infant.

Problems of Chap. 9

9.6 (partial answer) 4,720 for $z = 0$ (turbulent), 0.18 for $z = 20$ (laminar).

9.16 0.0078 cmH_2O/(L/s), which is much smaller than the total resistance of $\sim$2 cmH_2O/(L/s).

9.20 2.7 cmH_2O-s/L.

9.25 (a) Different people have different maximum expiratory pressures, and only a fraction of them could attain this maximum inflation pressure.

9.26 About half.

9.27 (a) For men $\simeq$1.5 m; (b) For an inspiratory volume of 1.8 L, the cross section is 10 cm^2, and diameter is about 3.9 cm (if this were the actual limitating factor).

9.30 Larger in a mouse (0.005 L/kg-cm-H_2O) than in man (0.003 L/kg-cm-H_2O).

9.37 (partial answer) 570 L of O_2 (at 1 atmosphere oxygen pressure) are consumed per day.

9.39 (a) Most of water vapor in the expired air is lost.
(b) Yes, more water vapor would be recovered, but you should not let the recovered water, in the form of mucus, leave your nose.

9.42 (partial answer) (a) 17.3 m^3, 8.5 ft.
(b) It decreases by 5.0% to 15.9%; the sailor starts feeling some effects due to this lower O_2 level.
(c) The volume would not change.
(d) The volume decreases by a factor of 4 here. The smaller the boat, the smaller the possible bubble, both at sea level and at greater depths.
(e) It increases by 5.0% (the same fraction as the O_2 fraction decreases here), to 5.04% (including the initial 400 ppm level); if the CO_2 does not dissolve in the water (it does to some degree), the sailor appears to be in great danger and greater danger from the high CO_2 levels than the low O_2 levels.

9.45 (partial answer) 225 mmHg total, 42 mmHg oxygen.

Problems of Chap. 10

10.4 Respectively at 0, 20, and 25 °C, v_{air} is 331, 343, and 346 m/s; ρ_{air} is 1.292, 1.204, and 1.184 kg/m^3; and Z_{air} is 428, 413, and 410 kg/m^2-s.

10.6 0.00002–2000 dyne/cm^2.

10.11 70 dB SPL.

10.14 (a) 100 dB SPL, 40 dB SPL.

10.20 (a) 0.27.

10.33 ∼1,000 Hz, which makes sense since the voices of children are higher pitched than those of adults.

10.38 160 Hz, assuming a mass with the mass of the vocal folds is attached to a massless spring with the force constant of the vocal folds. This really requires the analysis of a freely oscillating massive spring, which shows that oscillation frequency is $\pi/2 \times$ this value [691].

10.40 Lower, because they have higher fundamental buzzing frequencies.

10.47 3×10^{-13} m.

10.49 It is 9×10^6 larger at 3,000 Hz than that at 1 Hz.

10.55 (a) Radii of 0.4 μm for 20 Hz to 400 μm for 20 kHz.

10.60 17,000 Hz would be best because the auditory sensitivity of older people is very low at this frequency relative to that of younger people, at 250 Hz and 1,000 Hz the auditory sensitivity is not that different for older and younger people, at 30,000 Hz humans have no auditory sensitivity.

10.61 About 200/s.

10.66 100–8,000 Hz, over 40 dB.

10.67 40–14,000 Hz, over 70 dB.

10.72 The former (60 dB SPL) is a bit louder than the latter (59 dB SPL).

Problems of Chap. 11

11.1 (b) −2.5 D, 5.0 D, 7.5 D, 0.13 m.

11.14 (a) (partial answer) 2.5% for the first surface, (b) no.

11.20 7.51 mm, −7.51 mm.

11.21 6.07 mm.

11.23 The retina.

11.28 More damage is done if you look in the direction of the beam because it will focus on the fovea. Damage to the fovea can hurt sharp vision, leaving you with fuzzy vision.

11.29 8.8 cm.

11.40 (partial answer) Refractive index is larger at 630 nm by 0.0008 (if everything else is the same).

11.48 58.62 D (smaller than before), 22.8 mm (longer), mostly due to the smaller refractive index of the crystalline lens.

11.55 +3.72 *D*

11.56 −1.71 D, −1.67 D.

11.63 (c) Myopia. A correction of −2 D would lead to good vision. (The patient has 4 D of accommodation, which is sufficient with this correction.)

11.69 (b) 7.55 mm = 0.00755 m.

11.78 11.5°

11.79 You would use prism corrections in both eyes with the same prism power, with the grooves vertical in both cases. That for the left eye would look like that in Fig. 11.50 with the eye looking along the page starting from the bottom of the figure; it would be the (left-right) mirror image of this for the right eye.

11.81 250 lux assuming 500 lumens/W (Fig. 11.51). It is consistent with the levels given in Table 11.5.

Problems of Chap. 12

12.2 110,000 ohms.

12.4 (a) 240 mA, (b) shock and possible ventricular fibrilation would result.

12.11 The large net diffusion of K^+ outside, the impermeability of the membrane to the proteins, which are negatively charged, and the Na^+ pump contribute to the cell being negative relative to the extracellular fluid for a resting axon. The small net diffusion of Na^+ into the cell adds slightly to the positivity inside the cell.

12.19 (partial answer) 6.20×10^{-4} s for unmyelinated axons.

12.28 ~80/min.

12.30 The dipole usually rotates also. A(c); B(a); C(d); D(b).

Problems of Chap. 13

13.2 Every 17 min.

13.3 (partial answer) 1.6/s for a 1 s delay.